

Native implementation of the machine-learned Skala exchange–correlation functional in CP2K: Unified one-centre reconstruction for molecular and condensed-phase calculations

Johann Pototschnig^{1,2}, Franz Pöschel^{1,2}, Jürg Hutter³, and Thomas D. Kühne^{†1,2,4}

¹Center for Advanced Systems Understanding (CASUS), Görlitz, Germany

²Helmholtz-Zentrum Dresden-Rossendorf, Dresden, Germany

³Department of Chemistry, University of Zurich, Zurich, Switzerland

⁴Institute of Artificial Intelligence, Technische Universität Dresden, Dresden, Germany

[†]E-mail: tkuehne@cp2k.org

Abstract

Machine-learned exchange–correlation (XC) functionals combine high molecular accuracy with the prospect of efficient condensed-phase simulation. We implement Skala natively in CP2K using its Gaussian and plane-wave (GPW) and Gaussian and augmented-plane-wave (GAPW) methods. The central development is a joint one-centre reconstruction of density, density gradients, and kinetic-energy density before functional evaluation. This preserves mixed gradient terms and nonlocal couplings across the smooth and atom-local contributions for all-electron and pseudopotential calculations. An XC-specific GAPW representation separates rapidly varying local structure from the smooth plane-wave grid and also provides an efficient default starting point for calculations using pseudopotentials or effective core potentials. The implementation incorporates Brillouin-zone sampling and point-group symmetry reduction. On 70 common dietGMTKN55 reactions, mixed all-electron/pseudopotential calculations yield a mean absolute error of 1.89 kcal mol^{−1} and a median of 0.55 kcal mol^{−1}. On a 65-reaction pseudopotential intersection, the native and earlier GauXC routes give closely similar mean absolute errors of 2.22 and 2.18 kcal mol^{−1}. Crystalline CO₂, NH₃, and urea probe complementary interaction regimes, ranging from dispersion and quadrupolar electrostatics to hydrogen bonding and cooperative hydrogen-bond networks. With quadruple-zeta all-electron bases, Skala with D3(BJ) gives excellent agreement with diffusion Monte Carlo for this three-crystal set, with a mean absolute error of 1.54 kJ mol^{−1} and individual deviations of 2.67, 0.02, and 1.93 kJ mol^{−1}, respectively. Matched basis tests show that the initially stronger binding of CO₂ and urea has a substantial finite-basis contribution. The larger-basis results compare favourably with established dispersion-inclusive functionals. Numerical controls support the consistency of the native implementation and separate basis sensitivity from functional performance. The resulting framework connects molecular Skala to condensed-phase electronic structure and enables future development using periodic many-body reference data.

1 Introduction

The accuracy of Kohn–Sham density-functional theory (DFT) is governed to a large extent by the approximation to exchange and correlation (XC).^{1,2} Learning this contribution from accurate electronic-structure data offers a route beyond conventional functional forms. Skala combines local density information with a learned nonlocal representation and has demonstrated strong performance across molecular energetics.³ Its reported training data are molecular, including reaction energies and noncovalent interactions. Applying the same model to condensed phases therefore raises two distinct questions: can the functional be evaluated consistently within a periodic electronic-structure method, and does its molecular accuracy transfer to the resulting electronic environments?

A condensed phase is not simply a gas-phase calculation in a larger box. Coordination, polarization, screening, and collective response change the electronic environments sampled by the functional. Anderson’s “More Is Different”

perspective captures why knowledge of the constituents alone need not determine the behaviour of an extended system.⁴ For a learned density functional, the corresponding challenge is one of transfer between physical environments: the exact functional is universal, but a finite training set need not teach an approximation the collective response of condensed matter.

Our molecular implementation of Skala in CP2K through GauXC provides an independent reference for this development.^{5,6} There, a Gaussian-basis density matrix is evaluated on an atom-centred quadrature. Here, the interface instead uses density fields constructed by CP2K’s own Gaussian and plane-wave (GPW) and Gaussian and augmented-plane-wave (GAPW) machinery.^{7–9} This gives direct access to the existing treatment of boundary conditions, reciprocal-space electrostatics, Brillouin-zone sampling, and symmetry. The important methodological issue is not merely replacing one quadrature library with another. A learned nonlocal functional must receive a jointly reconstructed density, including

the one-centre contributions of the augmented representation.

We develop this reconstruction for all-electron (AE) and pseudopotential calculations and describe its variational coupling to the electronic degrees of freedom. A molecular comparison establishes the connection to the existing Skala implementations. We then consider three compact molecular crystals with complementary bonding motifs and high-level lattice-energy references. Detailed numerical tests and a ten-solid equation-of-state comparison are collected in the Electronic Supplementary Information (ESI). The ESI notation and scope are defined in Sec. S1. The aim is to establish a practical native implementation while making the distinction between implementation consistency and condensed-phase model accuracy explicit.

2 Theory and Implementation

2.1 From the density matrix to primitive fields

In a Gaussian basis, the spin density at a point is obtained from the occupied Bloch orbitals or, equivalently, the density matrix,

$$\rho_{\sigma}(\mathbf{r}) = \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{\mu\nu} P_{\mu\nu}^{\sigma}(\mathbf{k}) \chi_{\mu\mathbf{k}}(\mathbf{r}) \chi_{\nu\mathbf{k}}^*(\mathbf{r}). \quad (1)$$

The $\mathbf{k}$ -point weights are normalized, and $P^{\sigma}(\mathbf{k})$ is complex Hermitian, whereas the contracted density is real. The primitive input to the native interface is

$$\begin{aligned} \mathbf{u}_{\sigma}(\mathbf{r}) &= (\rho_{\sigma}, \nabla \rho_{\sigma}, \tau_{\sigma}), \\ \tau_{\sigma} &= \frac{1}{2} \sum_{n\mathbf{k}} w_{\mathbf{k}} f_{n\mathbf{k}\sigma} |\nabla \psi_{n\mathbf{k}\sigma}|^2. \end{aligned} \quad (2)$$

Thus Skala receives densities, gradients, and kinetic-energy densities, not the density matrix itself. The density matrix remains within CP2K, where the fields and the corresponding potential matrix are constructed.

On quadrature points $\mathbf{r}_i$ with weights W_i , the model energy has the form³

$$E_{\text{xc}}^{\theta} = -C_x \sum_i W_i \left(\rho_{\alpha i}^{4/3} + \rho_{\beta i}^{4/3} \right) f_{\theta}[\mathbf{x}]_i. \quad (3)$$

Here $C_x = \frac{3}{4}(6/\pi)^{1/3}$ and $\mathbf{x}$ contains spin densities, gradient invariants, and kinetic-energy densities. The enhancement factor f_{θ} depends on features at multiple points within an atom-associated block, rather than only at point i . Consequently, both the reconstructed fields and their spatial organization enter the discrete functional. Our contribution is the consistent construction and differentiation of this functional within GPW/GAPW, not a modification or retraining of the Skala model.

2.2 GPW and the augmented density decomposition

In GPW, Gaussian orbital products are collocated on a hierarchy of regular real-space grids and represented by plane waves for operations such as solving the Poisson equation.⁷ A finer plane-wave grid resolves more rapidly varying products, but increases the cost throughout the cell. This is particularly restrictive for the sharply varying core density of an AE

calculation. GAPW separates the global smooth density from atom-local contributions so that these two length scales need not be resolved on the same grid.⁸

For the primitive fields this separation is written

$$\mathbf{u}_{\sigma} = \tilde{\mathbf{u}}_{\sigma} + \sum_{A,\mathbf{L}} \left(\mathbf{u}_{A\mathbf{L},\sigma}^h - \mathbf{u}_{A\mathbf{L},\sigma}^s \right), \quad (4)$$

where A labels atoms and $\mathbf{L}$ the periodic images required by the boundary conditions. The hard field restores the local representation, while the soft field subtracts the corresponding contribution already included in the global smooth field. The subtraction prevents double counting: a one-centre correction is not an independent atomic energy added to the system.

The same construction is needed for all components of $\mathbf{u}_{\sigma}$. For example, in a real atom-local basis $\{\phi_{Aa}^q\}$, with $q = h, s$, the one-centre density and kinetic-energy density have the structure

$$\begin{aligned} \rho_{A\sigma}^q(\mathbf{r}) &= \sum_{ab} D_{A,ab}^{q,\sigma} \phi_{Aa}^q(\mathbf{r}) \phi_{Ab}^q(\mathbf{r}), \\ \tau_{A\sigma}^q(\mathbf{r}) &= \frac{1}{2} \sum_{ab} D_{A,ab}^{q,\sigma} \nabla \phi_{Aa}^q(\mathbf{r}) \cdot \nabla \phi_{Ab}^q(\mathbf{r}). \end{aligned} \quad (5)$$

The local coefficients $D_{A,ab}^{q,\sigma}$ are obtained from the electronic density matrix through the GAPW projection. Density gradients follow by differentiating the orbital products. Reconstructing ρ alone is therefore insufficient: τ is an orbital-dependent field and cannot be inferred by differentiating the reconstructed density. Finite one-centre expansions and finite grids introduce representation and integration errors into Eq. (4). Their convergence must be checked together with the energy and its derivatives.

2.3 AE and pseudopotential GAPW

For AE GAPW (GAPW-AE), this reconstruction contains the explicitly treated core and valence electrons. With a Goedecker–Teter–Hutter (GTH) effective-core potential (ECP), it reconstructs the explicit valence representation without reintroducing core electrons removed by the effective-core Hamiltonian.^{10;11} This distinction is essential when comparing AE and pseudopotential results. Their difference is not, in general, an error of numerical integration.

In the AE route, the hard-minus-soft contribution is intrinsic to the chosen GAPW representation. Passing only its smooth part to Skala would omit part of the AE fields. An AE model evaluation without an *explicit* one-centre correction is nevertheless possible in a different construction: direct evaluation of the complete Gaussian-basis density matrix on an adequately resolved atom grid, as in the molecular GauXC interface. There the full fields are already present. The need for reconstruction is thus determined by how the fields are represented, not by whether the integration library is GauXC or native CP2K.

For pseudopotential GAPW we distinguish a direct-valence route from an augmented-valence route. The former supplies the direct regular-grid valence fields. The latter includes the atom-local hard-minus-soft fields in Eq. (4). The local reconstruction is analogous to the projector augmented-wave (PAW) method in its algebraic organization, but it is neither a change to a PAW Hamiltonian nor an AE reconstruction of

the removed core. The same ECP and explicit electron count are retained. At the chosen finite representation, the two routes need not present identical fields to the nonlinear model. Their agreement must therefore be assessed for observables rather than inferred solely from a common pseudopotential.

The selection of one-centre contributions is made by atomic kind. A mixed AE/ECP calculation consequently retains the AE corrections even when the ECP kinds use direct-valence fields. In the molecular benchmark, “with/without one-centre correction” refers to this ECP choice, not to removing the intrinsic AE reconstruction. For the light-element molecular crystals, AE and all-ECP calculations instead provide separate, complete core-treatment comparisons.

2.4 The XC-specific GAPW representation

The XC-specific GAPW representation (GAPW-XC) applies the augmented construction specifically to the XC fields. The electrostatic and XC representations can then use different smooth densities derived from the same orbital density matrix. CP2K collocates a separate, softer XC density, denoted $\tilde{\rho}_{xc}$, together with the corresponding gradient and kinetic-energy density. The complementary atom-local fields restore the structure shifted out of this regular-grid contribution. In compact form,

$$\mathbf{u}_{\sigma}^{XC} = \tilde{\mathbf{u}}_{xc,\sigma} + \sum_{A,L} (\mathbf{u}_{AL,\sigma}^h - \mathbf{u}_{AL,\sigma}^s). \quad (6)$$

The associated hard/soft projections and smooth collocation must be used consistently. The smooth term cannot simply be exchanged while leaving its reverse mapping unchanged. This XC-specific decomposition does not change the selected GTH potential or add core electrons.

The practical advantage is that rapidly varying local XC structure can be resolved on radial-angular grids without forcing an equally fine plane-wave grid over the whole cell. GAPW-XC is therefore our recommended default starting representation for GTH/ECP native Skala calculations. The paired conformational cutoff study in ESI Sec. S3.3 supports its faster convergence relative to GPW. The one-centre reconstruction makes this efficiency possible while retaining the model’s response to the complete augmented fields. Standard GAPW and GAPW-XC distribute the regular-grid and atom-local work differently, so their finite-resolution errors are assessed through complete energy differences and derivatives. This numerical advantage is separate from the physical approximation introduced by replacing core electrons with a pseudopotential.

Table 1 separates the electronic Hamiltonian, the smooth XC field, and the local reconstruction. Merely deleting the matching local terms from Eq. (6) would leave the softer XC field uncompleted. This must be distinguished from selecting the direct-valence ECP route: that choice also retains the unsmoothed valence basis for field construction and therefore supplies GPW-like XC fields, even within a calculation otherwise labelled GAPW-XC. It is a complete alternative representation, not an incomplete augmented one. In a standard GAPW calculation, choosing direct-valence XC likewise does not by itself replace the remaining electrostatic machinery.

2.5 One reconstruction, one joint functional

Classical GAPW organizes local or semilocal XC integration as a smooth-grid contribution plus atom-centred hard-minus-soft energy integrals. Its validity relies on the construction and matching of the local fields, not on linearity of the XC functional. For a nonlinear nonlocal model, the expression

$$E_{xc}[\tilde{\mathbf{u}}] + \sum_A (E_{xc}[\mathbf{u}_A^h] - E_{xc}[\mathbf{u}_A^s]) \quad (7)$$

is not interchangeable with evaluating the functional on Eq. (4). The model is nonlinear and nonlocal. Separate evaluations omit both mixed local descriptors and couplings between different contributions to the reconstructed environment.

Even a quadratic gradient invariant illustrates the issue. With $\mathbf{g} = \tilde{\mathbf{g}} + \mathbf{g}^h - \mathbf{g}^s$,

$$\begin{aligned} |\mathbf{g}|^2 &= |\tilde{\mathbf{g}}|^2 + |\mathbf{g}^h|^2 + |\mathbf{g}^s|^2 + 2\tilde{\mathbf{g}} \cdot \mathbf{g}^h \\ &\quad - 2\tilde{\mathbf{g}} \cdot \mathbf{g}^s - 2\mathbf{g}^h \cdot \mathbf{g}^s. \end{aligned} \quad (8)$$

The cross terms exist wherever the corresponding gradient fields overlap, particularly in augmentation regions. Notice also the positive $|\mathbf{g}^s|^2$ term in the reconstructed invariant: subtracting a separately evaluated soft energy is not equivalent to subtracting a field before forming its nonlinear descriptors. The spin-coupled case follows the same principle. The six smooth, hard, and soft spin-gradient vectors admit fifteen unordered off-diagonal products, twelve of which couple different components. The full expansion is given in ESI Sec. S2.1. A larger common grid alone does not recover these terms if the three densities are still passed to separate functional evaluations.

The nonlocal difficulty goes beyond this local algebra. A hard contribution near one centre can alter the learned response at another point in the same descriptor block, where the smooth density and neighbouring augmentation fields also contribute. Separate hard and soft model evaluations never see that combined environment. Even exact quadrature of the separated expressions would not restore these missing dependencies.

We therefore interpolate the smooth fields and add the hard-minus-soft fields *pointwise*, before forming model descriptors. Skala then evaluates the jointly reconstructed fields. Here, “one evaluation” means one consistent functional of the combined fields, not a restriction to one physical inference call: atom-block batching and parallel execution remain possible. The distinction is between computational partitioning of the same functional and replacing it by a sum of different functionals.

2.6 Accurate one-centre integration and joint reconstruction

The classical accurate GAPW XC integration scheme addresses a different difficulty: cancellation between smooth-grid and soft atom-centred integrals evaluated on dissimilar quadratures.^{8,12} Using an atom-local switch

$$s_A(\mathbf{r}) = \exp[-\alpha_A |\mathbf{r} - \mathbf{R}_A|^2], \quad (9)$$

Table 1: Density representations in the native interface. One-centre fields are added before descriptor formation and Skala evaluation. No removed core electrons are restored by hard-minus-soft valence reconstruction.

Representation	Hamiltonian	Smooth XC field	Atom-local contribution
GPW	ECP	Direct valence	None
GAPW-AE	AE	Smooth GAPW	Hard minus soft, including explicit core and valence
GAPW-GTH, direct	ECP	Direct valence	None for ECP kinds
GAPW-GTH, one-centre	ECP	Smooth GAPW	Hard-minus-soft valence fields
GAPW-XC/GTH	ECP	Softer XC-specific field	Matching hard-minus-soft valence fields

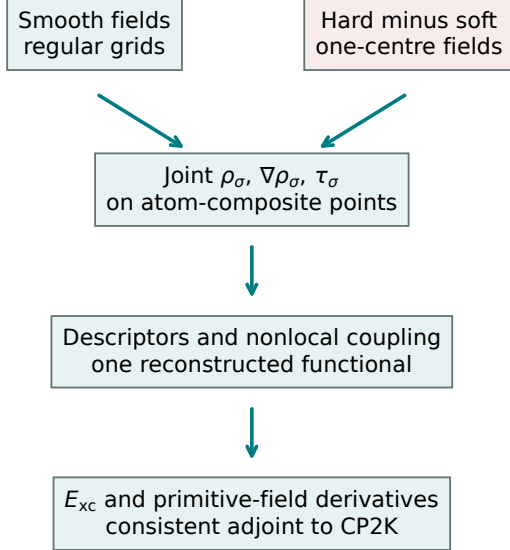


Figure 1: The native interface reconstructs the primitive fields before descriptor formation and Skala evaluation. The reverse operation follows the adjoint of the same reconstruction, preserving the hard-minus-soft signs. The atom-centred quadrature resolves local structure without requiring a uniformly fine regular grid.

with periodic images understood, its local/semilocal energy organization is schematically

$$E_{xc}^{acc} \simeq \int \left(1 - \sum_A s_A \right) e_{xc}[\tilde{u}] d\mathbf{r} + \sum_A \int \{ e_{xc}[\mathbf{u}_A^h] - (1 - s_A) e_{xc}[\mathbf{u}_A^s] \} d\mathbf{r}. \quad (10)$$

Here e_{xc} is a pointwise energy density, not an independent nonlocal Skala calculation. Near a nucleus, the hard atom-centred term carries the rapidly varying contribution while the nearly cancelling smooth and soft terms are suppressed. The local matching assumptions of GAPW underlie this rearrangement. The switching weights, quadrature, and their geometric derivatives must remain consistent.

Equation (10) does not supply the mixed descriptors missing from Eq. (7). The native Skala term instead uses its own jointly reconstructed fields and atom-composite quadrature. We retain accurate integration for the applicable classical GAPW terms, but do not multiply the Skala reconstruction by the classical switching weights or add separately evaluated

hard/soft Skala energies to it. This distinction separates an improvement of classical integration from the extension required for a learned nonlocal functional.

2.7 Periodic atom-composite quadrature

For a radial–angular point $\mathbf{r}_{Ai}$ centred on atom A , the discrete reconstruction is

$$\mathbf{u}_{Ai,\sigma} = \sum_g I_{Ai,g} \tilde{\mathbf{u}}_{g,\sigma} + \sum_{B,L} [\mathbf{u}_{BL,\sigma}^h(\mathbf{r}_{Ai}) - \mathbf{u}_{BL,\sigma}^s(\mathbf{r}_{Ai})]. \quad (11)$$

The interpolation I maps the regular-grid fields to the target atom grid. The second sum includes neighbouring centres and periodic images, not only the target atom. Thus “one-centre” describes the source expansion. It does not mean that its contribution is confined to a separate model call or omitted from neighbouring descriptor blocks. The sum is absent for direct-valence ECP kinds, while intrinsic AE contributions are retained.

Let w_{Ai} denote the unpartitioned radial–angular quadrature. Smooth atom-image partition functions define the energy weights,

$$p_{AL}(\mathbf{r}) = \frac{q_{AL}(\mathbf{r})}{\sum_{B,L'} q_{BL'}(\mathbf{r})}, \quad \sum_{A,L} p_{AL}(\mathbf{r}) = 1, \quad (12)$$

so that a target grid carries $W_{Ai} = w_{Ai} p_{A0}(\mathbf{r}_{Ai})$. The differentiable shape functions q provide a Becke-like partition.¹³ Periodic images distribute the integration over equivalent cells without double counting. A distinct partition over the target atom’s own periodic images defines its nonlocal descriptor domain. This self-image partition does not cut that domain at boundaries between different atoms. Energy partitioning and descriptor weighting therefore have different roles.

The same construction reduces to a molecular atom-composite quadrature when no periodic images are required. A regular common grid can also support joint reconstruction, but resolving hard local fields everywhere is expensive. On the atom-composite grid, radial and angular resolution control the local structure independently of the plane-wave cutoff for the smooth contribution. This is an efficiency distinction, not a reason to evaluate different functionals on the two layouts.

Particle number provides an additional consistency check. The explicit electronic count is determined by the occupied density matrix and overlap matrix,

$$N = \sum_{\mathbf{k}\sigma} w_{\mathbf{k}} \text{Tr}[\mathbf{P}^\sigma(\mathbf{k}) \mathbf{S}(\mathbf{k})]. \quad (13)$$

The integral of the represented density is organized as $\tilde{N} + \sum_A (N_A^h - N_A^s)$, whereas the finite model quadrature gives $N_Q = \sum_{Ai\sigma} W_{Ai} \rho_{Ai,\sigma}$. A discrepancy between N_Q and Eq. (13) tests the field representation and quadrature. It does not, by itself, imply a change in the self-consistent-field (SCF) occupations. No density-dependent renormalization is applied, since it would alter both the functional and its variational derivative. Cutoff, radial/angular, and one-centre convergence are consequently distinct from changing the core treatment.

Finally, a nonlinear core correction (NLCC) must not be confused with hard-minus-soft reconstruction. An NLCC potential supplies a separate frozen-core density that can be added to the valence density and gradient before descriptor formation. It does not by itself provide a corresponding core kinetic-energy density. This is not equivalent to the explicit core orbitals of GAPW-AE, and it is not the origin of the one-centre corrections compared here.

2.8 Variational coupling, forces, and stress

Let $\mathcal{R}$ denote the discrete reconstruction from the density matrix P to the model fields. The XC contribution to the generalized Kohn–Sham matrix is obtained by the chain rule,

$$V_{xc} = \left(\frac{\partial \mathcal{R}}{\partial P} \right)^* \frac{\partial E_{xc}}{\partial \mathbf{u}}. \quad (14)$$

The asterisk denotes the adjoint under the chosen discrete contractions. With $\bar{\mathbf{u}}_{Ai} = \partial E_{xc} / \partial \mathbf{u}_{Ai}$, Eq. (11) gives, component by component,

$$\begin{aligned} \bar{\mathbf{u}}_g &= \sum_{Ai} I_{Ai,g} \bar{\mathbf{u}}_{Ai}, \\ \bar{D}_A^h &= \left(\frac{\partial \mathbf{u}_A^h}{\partial D_A^h} \right)^* \bar{\mathbf{u}}, \quad \bar{D}_A^s = - \left(\frac{\partial \mathbf{u}_A^s}{\partial D_A^s} \right)^* \bar{\mathbf{u}}. \end{aligned} \quad (15)$$

ESI Sec. S2.2 details the discrete adjoint and moving-grid terms. The contractions include every target point to which a source contributes. The quadrature weights are already included in $\bar{\mathbf{u}}$, so applying them a second time would be incorrect. The local coefficients are then projected back to the orbital density matrix. Gradient and kinetic-energy-density adjoints use the differentiated Gaussian products in Eq. (5). Thus the reverse path is the transpose of the actual discrete forward map, rather than an independently interpolated potential. This consistency is particularly important for the cancellation of hard and soft contributions and for self-consistency close to a numerical floor.

At fixed orbital density matrix, the explicit XC derivative with respect to a nuclear coordinate or cell-strain parameter λ includes geometry-dependent mappings,

$$\left. \frac{\partial E_{xc}}{\partial \lambda} \right|_P = \sum_{Ai} \bar{\mathbf{u}}_{Ai} \cdot \left. \frac{\partial \mathbf{u}_{Ai}}{\partial \lambda} \right|_P + \left. \frac{\partial E_{xc}}{\partial \lambda} \right|_{\mathbf{u}}. \quad (16)$$

The first term includes interpolation and hard/soft-field derivatives, while the second includes explicit model-coordinate and quadrature-weight dependence, including both periodic partitions. Stationarity of the complete electronic energy and the basis/overlap Pulay terms are then used to obtain the total force and stress. The XC contribution alone is not

stationary with respect to the density matrix. Model differentiation alone is therefore not a complete derivative of the self-consistent total energy. The nuclear force is $-dE/d\mathbf{R}_A$, and the strain derivative per cell volume gives the stress in the adopted sign convention. Self-consistent finite-difference (FD) checks of a force and diagonal stress component in a periodic water test support the derivative consistency of all five representations. The FD protocol, numerical errors, and tested scope are documented in ESI Sec. S2.3 and Table S1.

2.9 K-points, symmetry reduction and parallelism

Brillouin-zone integration enters through the density matrix and Eq. (1). Symmetry reduction must transform orbital components and Bloch phases consistently, together with the integration weights. Multiplying a scalar result by a symmetry count is insufficient. Once the real primitive fields are assembled, Skala need not know whether they originated from a full k mesh or a symmetry-reduced one. Their adjoints are returned to the corresponding complex Hermitian Kohn–Sham matrices. The native implementation uses CP2K’s k-point and symmetry infrastructure,^{14;15} with full point-group reduction in the molecular-crystal calculations and numerical full/reduced-mesh comparisons in ESI Sec. S5 and Table S11.

CP2K supplies fields, atom-associated coordinates, energy weights, and descriptor weights directly to the same Skala TorchScript model used by the molecular interface. LibTorch returns the energy and input derivatives without a GauXC integration step. Complete atom blocks can be distributed over Message Passing Interface (MPI) ranks or batched on a device because the nonlocal processing retains the atom-block structure. Threaded field construction and back-projection complement inference on central processing units (CPUs) or graphics processing units (GPUs). Splitting the computational batch preserves the functional only when each block’s descriptor domain and all contributing reconstructed fields are retained. This provides parallel execution without reverting to the separated smooth/hard/soft functional in Eq. (7).

3 Computational Strategy

We use Skala-1.1 with its accompanying D3(BJ) dispersion contribution.^{3;16;17} All comparisons retain the same model and dispersion convention within a dataset. We report the mean absolute error (MAE) and complementary error statistics. The Gaussian bases and GTH potentials follow the University of Zurich (UZH) 2026.2 definitions.¹⁸ The molecular study uses triple-zeta AE bases where available and pseudopotential bases for the heavy-element part of the mixed protocols. “Mixed” refers to core treatment, not to a hybrid XC functional.

For molecules, the comparison uses 70 common reactions from dietGMTKN55^{19;20} with complete energies in all three native protocols: all-GTH GAPW-XC and mixed GAPW-AE/GTH with direct (D) or one-centre (1C) GTH fields, labelled Mixed D and Mixed 1C in the tables. The remaining cases required longer SCF convergence and are not included. The same reaction set and dispersion contribution

are used across these native protocols, with errors measured against the official reference values. Comparisons with earlier molecular implementations retain each source’s recorded dispersion contribution. ESI Secs. S3 and S4 give the basis, integration, and selection details, including the basis/core protocol (Sec. S3.1) and separate cutoff, radial/angular, and cell-padding studies (Secs. S3.3–S3.5).

ESI Sec. S6.1 specifies the molecular-crystal protocol. The condensed-phase energy study uses CO₂, NH₃, and urea geometries from the versioned supporting dataset of Della Pia *et al.*²¹ The corresponding electronic reference energies are from diffusion Monte Carlo (DMC). The systems contain only light elements, allowing a direct AE comparison without mixed-core substitutions. All four representations are complete for both crystal and molecule. The electronic lattice energy is

$$E_{\text{latt}} = E_{\text{crystal}}/Z - E_{\text{molecule}}, \quad (17)$$

where Z is the number of molecules in the cell and negative values denote binding. The gas-phase energy uses the independently supplied molecular reference geometry. The comparison is between electronic energies, without thermal or zero-point contributions.

Crystal and molecular energies use matched basis and quadrature definitions, with nonperiodic electrostatics for the gas-phase references. All three reported AE pairs use the UZH quadruple-zeta valence plus double-polarization (QZVPP) basis and a common finer atom quadrature, following the paired tests in ESI Sec. S6.3. The GTH results retain their original basis and quadrature. The crystals employ Gamma-centred 3³ k meshes with full point-group reduction. ESI Secs. S6.1 and S6.2 document the production settings, basis and grid choices, cutoff and k-point sensitivities, and symmetry comparisons. Self-consistent FD force and stress tests are given in Sec. S2.3. These controls establish the numerical basis for the following comparison without changing the functional or fitting it to the reference energies.

4 Results and Discussion

4.1 Molecular energetics and the GauXC connection

Table 2 summarizes the 70 common reactions. The mixed AE/pseudopotential protocols have an MAE of 1.89 kcal mol^{−1}, compared with 3.51 kcal mol^{−1} for all-GTH GAPW-XC. Their median absolute errors, however, are similar: 0.55 and 0.60 kcal mol^{−1}. Most reactions are consequently described much better than the all-GTH aggregate MAE alone would suggest. Absolute errors are below 1 kcal mol^{−1} for 46 of 70 GAPW-XC reactions and 42 of 70 mixed-protocol reactions.

The similar medians show that the larger all-GTH MAE is driven by a small number of core-treatment-sensitive reactions. The largest is the ozone-containing W4-11/138 reaction, whose signed error changes from −68.49 kcal mol^{−1} with all-GTH to +9.27 kcal mol^{−1} with mixed core treatment. It remains in the statistics. This contrast separates the typical molecular accuracy from sensitivity to the core Hamiltonian and basis. The complete reaction-resolved comparison is given in ESI Sec. S4 and Table S6.

Table 2: Errors against official references for the same 70 dietGMTKN55 reactions, in kcal mol^{−1}. The two mixed protocols share the same AE results for the light-element part.

Statistic	GAPW-XC/GTH	Mixed D	Mixed 1C
Mean signed error	−2.439	0.535	0.535
MAE	3.505	1.890	1.891
Root-mean-square error	10.675	3.650	3.650
Median absolute error	0.603	0.555	0.555
Maximum absolute error	68.488	20.450	20.450

The direct and one-centre mixed protocols differ by at most 0.001915 kcal mol^{−1} in this set, where the light-element AE energies are shared. The additional GTH reconstruction therefore makes little difference to these retained reaction energies. Its methodological role is nevertheless essential: it supplies the complete augmented fields to the nonlinear functional, irrespective of the extent to which its contribution cancels in an energy difference. The molecular-crystal comparison below tests that role independently in all-GTH systems. Targeted paired AE basis controls in ESI Sec. S3.2 demonstrate that some molecular reaction energies also retain substantial finite-basis sensitivity. Additional controls reduce the chlorine-dimer electron-affinity error from 2.97 to 0.28 kcal mol^{−1} with QZVPP, whereas the error in a hydrogen-bonded complex increases from 0.17 to 0.54 kcal mol^{−1} (ESI Table S3). Thus the improvement is not uniform across reactions. Associated restart controls identify a higher-energy SCF solution behind the initially anomalous QZVPP ethane torsional barrier. These tests are reported separately and do not alter the uniform 70-reaction statistics in Table 2.

The earlier molecular GauXC implementation provides an independent point of comparison.^{5;22} Native GAPW-XC/GTH is compared with GauXC GPW-GTH, keeping both calculations in the pseudopotential regime. On their 65 shared reactions, the reference MAEs are closely similar at 2.219 and 2.181 kcal mol^{−1}, respectively. Their direct mean absolute difference (MAD) is 0.463 kcal mol^{−1}. The native route thus reaches the same benchmark accuracy scale without GauXC, despite using different Gaussian bases and GTH parametrizations. Together with the independent numerical consistency tests, this agreement validates the native implementation for the molecular applications considered here.

The reaction-resolved collaborator data also permit a separate comparison of the mixed native AE/GTH protocols with the earlier GauXC AE/def2-ECP and primary Microsoft PySCF/Skala results on all 70 retained reactions. The PySCF reference uses Becke partitioning without an atomic-radius adjustment. Here, the reference MAEs are 1.890/1.891 kcal mol^{−1} for the two native protocols, versus 1.238 for GauXC and 1.231 for PySCF. The native Mixed D–GauXC MAD is 1.889 kcal mol^{−1}, and the GauXC–PySCF MAD is 0.046 kcal mol^{−1}. The principal protocol distinction is the use of the UZH basis/core treatment in the native calculations, compared with def2 Gaussian bases and def2 ECPs in the earlier AE/ECP calculations. The remaining differences are consistent with these different finite-basis and core descriptions, together with the numerical integration settings. Differences between the recorded dispersion contri-

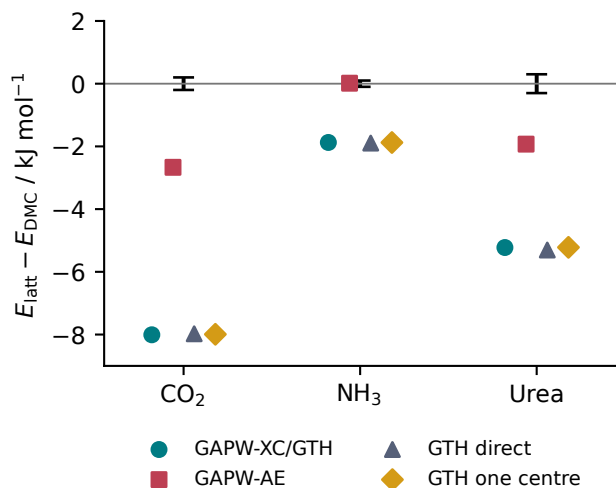


Figure 2: Signed Skala-DMC lattice-energy residuals. The three GTH routes nearly coincide. The quadruple-zeta AE results reduce the residuals, with ammonia close to zero. DMC statistical uncertainties are shown around zero. They do not include all reference systematic errors.

butions are much smaller. ESI Sec. S4.1 and Tables S7–S9 document both reaction-matched comparisons, individual energies, and dispersion sensitivity. The original 100-reaction GauXC/PySCF aggregates are retained only as context in Table S10.

4.2 Three molecular crystals with complementary bonding

CO₂, NH₃, and urea form a deliberately compact test of gas-to-condensed-phase transfer. CO₂ probes dispersion and quadrupolar electrostatics without hydrogen bonds. Ammonia adds a hydrogen-bonded molecular environment, while urea combines a polar carbonyl group with a cooperative, multiply connected hydrogen-bond network. The selection spans physically distinct interactions without making a claim to represent the full X23 distribution. Their small cells also permit direct comparison of AE and pseudopotential density representations.

The three GTH representations give stronger binding than DMC and agree closely with one another (Table 3 and Fig. 2). Their largest spread is 0.0853 kJ mol⁻¹, for urea. The one-centre-minus-direct shifts are -0.0159, +0.0182, and +0.0853 kJ mol⁻¹ for CO₂, NH₃, and urea. Thus the overbinding of the GTH calculations is not removed by switching among these three native density representations. The larger-basis AE results instead approach the DMC references, with essentially zero residual for ammonia.

The AE calculations bind less strongly than GAPW-XC/GTH by 5.34, 1.89, and 3.30 kJ mol⁻¹ for CO₂, NH₃, and urea, respectively. Their absolute DMC residuals are 2.67, 0.02, and 1.93 kJ mol⁻¹. Ammonia is the most accurate case, while CO₂ retains the largest absolute and fractional error in every representation. Enlarging both the crystal and molecular AE bases at fixed fine quadrature reduces the three-crystal MAE from 7.95 to 1.54 kJ mol⁻¹. The individual lattice energies weaken by 4.31, 0.56, and 14.42 kJ mol⁻¹, respectively. Thus the native implementation delivers highly

accurate molecular-crystal energies when sufficient AE basis flexibility is supplied, without changing the functional, geometries, or one-centre reconstruction. The paired triple-zeta results, SCF checks, and separate quadrature test are documented in ESI Sec. S6.3 and Tables S14 and S15.

The unequal basis requirements of AE and GTH calculations have a physical and a representational origin. An AE orbital basis must describe the tightly localized core orbitals and the nodal structure and environmental response of the valence orbitals, whereas GTH replaces the core electrons and smooths the valence orbitals. Moreover, the UZH AE bases are def2-derived bases optimized for GAPW using the molecularly optimized (MOLOPT) strategy, rather than unchanged def2 sets.^{18;23;24} Their finite-basis balance therefore need not match either the original def2 family or the independently optimized GTH valence bases at a similar nominal zeta level. For a lattice energy, the decisive quantity is the differential basis error between crystal and molecule. The present tests identify the larger AE basis as important for that balance, rather than suggesting a failure of the one-centre reconstruction.

The different interaction balances provide a plausible physical interpretation. Dispersion-dominated CO₂ tests a delicate balance of attractive correlation, quadrupolar electrostatics, and short-range repulsion.²⁵ Ammonia contains an N-H...N network, while urea combines N-H...O contacts with a polar carbonyl/amide framework and cooperative hydrogen bonding.²⁶ Earlier correlated calculations also find a larger gas-to-crystal intramolecular deformation contribution for urea.²⁵ Environment-dependent polarization, short-range interactions, and deformation can therefore contribute differently to the energy error. A monotonic explanation based only on molecular size or polarizability is insufficient.

These mechanisms suggest an interpretation rather than a decomposition of the Skala error. For CO₂, excessive attraction or insufficient short-range repulsion could each produce the remaining overbinding. Its smaller reference lattice energy also amplifies the fractional error: the 2.67 kJ mol⁻¹ AE discrepancy corresponds to approximately 9%, compared with about 27% for the GTH protocols. This contrasts with the consistently small NH₃ residual, which may involve favourable cancellation between crystal and molecular errors. Moreover, AE versus GTH changes the Hamiltonian, basis, and density together, whereas direct versus one-centre GTH isolates the reconstruction choice. The paired basis tests demonstrate why a large finite-basis AE-GTH difference should not be interpreted as evidence for an error in the one-centre reconstruction or as a direct measure of hydrogen-bond cooperativity.

The representative quadrature, k-point, symmetry, and CPU/GPU controls in ESI Sec. S6.2 and Table S13 are much smaller than the CO₂ reference discrepancy. Together with the agreement among GTH representations, they support a physical-protocol discrepancy rather than the tested integration or device effects. The energies remain finite-basis predictions. A complete-basis and counterpoise-corrected limit is outside the present comparison.

The comparison concerns the combined Skala-D3(BJ) protocol. Since dispersion is retained in every calculation, the residual cannot be uniquely decomposed into learned-XC and empirical-dispersion contributions. DMC provides a directly electronic many-body reference and avoids comparing a

Table 3: Electronic lattice energies in kJ mol^{-1} per molecule at the supplied geometries, including D3(BJ) for all Skala results. All AE entries use QZVPP, while GTH entries retain their molecularly optimized triple-zeta bases. The paired AE triple-zeta controls are reported in the ESI. DMC reference values are from Ref. 21. Parentheses give its statistical uncertainties, not a complete systematic uncertainty. Negative energies denote binding.

Crystal	GAPW-XC/GTH	GAPW-AE	GAPW-GTH direct	GAPW-GTH one centre	DMC
CO ₂	-37.408	-32.067	-37.376	-37.392	-29.4(2)
NH ₃	-40.074	-38.185	-40.094	-40.076	-38.2(1)
Urea	-113.725	-110.427	-113.805	-113.719	-108.5(3)

static electronic energy with an uncorrected sublimation enthalpy.^{21;27}

Published dispersion-inclusive DFT results place the material-dependent accuracy of the native calculations in context.²⁸ For NH₃, the native AE and GAPW-XC/GTH absolute errors of 0.02 and 1.87 kJ mol^{-1} compare favourably with 5.56 for PBE+D3 and 3.49 for SCAN+rVV10. For urea, GAPW-XC/GTH gives an error of 5.23 kJ mol^{-1} , close to 5.05 for SCAN+rVV10 and 4.65 for vdW-DF2. The larger-basis AE result reduces the Urea error to 1.93 kJ mol^{-1} , on the scale of the estimated overall DMC uncertainty. Across the three crystals, the AE MAE is 1.54 kJ mol^{-1} , compared with 3.76 for PBE+D3 and 3.54 for SCAN+rVV10. CO₂ retains the largest native AE residual, but its magnitude is substantially smaller after the basis enlargement. ESI Sec. S6.4 and Table S16 place all native representations alongside the DMC references and literature values. Small differences should be interpreted in light of the approximately 2 kJ mol^{-1} overall DMC uncertainty.²¹ Different geometries and numerical protocols, together with the small sample of three crystals, preclude a definitive ranking of the methods.

4.3 Condensed-phase transfer and the role of many-body data

The ten-solid structural analysis in ESI Sec. S8 provides complementary context. Its structure selection, fit stability, representation trends, and matched literature comparisons are detailed in Secs. S8.1–S8.5. Pressure derivatives and compressibilities derived from the same curves are reported in Sec. S9 and Table S30. Every retained native equilibrium lattice constant is shorter than the static-lattice experimental reference, with MAEs of 0.073–0.084 Å. Matched literature comparisons show smaller structural errors for established solid-state approximations and correlated methods.^{29;30} The uniform ten-solid comparison uses the UZH triple-zeta AE orbital basis. Additional matched five-volume QZVPP controls improve the lattice constant and bulk modulus of diamond but increase the contraction and stiffness of Si, while leaving the excessive contraction of MgO largely unchanged (ESI Sec. S8.3 and Table S19). Basis enlargement therefore has a material-dependent effect on these structural observables. The three controls do not establish basis convergence for all ten solids. An equation of state alone does not measure a cohesive energy, and its finite-basis structural errors should be distinguished from the molecular-crystal lattice-energy comparison.

The ice comparison provides a broader test of hydrogen-bonded condensed phases.³¹ On the same twelve phases at both basis levels, enlarging the AE basis reduces the absolute

lattice-energy MAE from 21.24 to 1.23 kJ mol^{-1} . The MAE of the eleven energy differences relative to Ih also decreases, from 4.09 to 0.77 kJ mol^{-1} . The latter comparison cancels the gas-phase reference exactly, showing that the improvement is not merely a common shift in binding. Small phase-ordering differences remain, and these fixed-geometry electronic energies do not determine finite-temperature phase stability. ESI Sec. S7 and Tables S17 and S18 give the phase-resolved energies, molecular-reference convention, and symmetry controls. Ice XIII required longer SCF convergence and is excluded from both basis-level averages.

Taken together, the larger-basis molecular-crystal results demonstrate accurate transfer to these condensed-phase environments. They also show why finite-basis effects must be separated from the performance of a learned functional. Skala was developed from molecular reference data rather than periodic-solid densities and energies,³ yet the present QZVPP results do not support a general failure for condensed matter. Extended coordination, screening, and cooperative polarization still motivate broader validation and training data across condensed-phase environments. The remaining material-dependent residuals and finite-basis contraction in the ten-solid set identify useful targets for that programme, rather than establishing the molecular training domain as their cause. A transferable approximation must capture the collective response of condensed matter as well as interactions among gas-phase molecules.

Periodic many-body calculations provide a concrete route to the necessary reference information. Booth, Grüneis, Kresse, and Alavi demonstrated highly accurate wavefunction treatments of real solids.³² Developments underlying CC4S make periodic coupled-cluster calculations increasingly accessible,^{33–35} and their use in machine-learned condensed-phase simulations is already practical.³⁶ Azadi, Booth, and Kühne combined periodic coupled cluster, DMC, and full configuration-interaction quantum Monte Carlo (FCIQMC) for the equation of state of atomic solid hydrogen.³⁷ Such complementary methods allow basis and finite-size errors to be assessed independently of a density-functional approximation.

Quantum Monte Carlo (QMC) offers flexible many-electron descriptions through neural-network trial states,^{38;39} antisymmetric shadow wavefunctions for solid hydrogen,⁴⁰ and resonating-valence-bond approaches in TurboRVB.⁴¹ Applications with Azadi span molecular ozone and correlated kagome systems, illustrating the reach from gas-phase to condensed-phase electronic structure.^{42;43} These methods are prospective sources of training and validation data, not ready-made Skala training sets. The relevant programme requires consistently converged energies and density information

across volumes, coordination environments, and distortions, together with self-consistent stability checks. The native implementation provides the computational framework in which that programme can be carried out.

5 Conclusions

The joint one-centre reconstruction brings learned nonlocal XC into CP2K’s native GPW/GAPW framework without GauXC. Reconstructing density, gradients, and kinetic-energy density before evaluation preserves the mixed terms and nonlocal couplings required by GAPW-XC, GAPW-AE, and augmented GAPW-GTH/ECP. GAPW-XC is the recommended default starting representation for pseudopotential calculations because local XC structure is resolved independently of the global plane-wave cutoff. The same variational construction connects the functional to CP2K’s forces, stress, k-point sampling, and point-group reduction, and to its boundary-condition framework for systems of different dimensionality. The present validation covers molecules and three-dimensional crystals.

The molecular results retain Skala’s high typical accuracy, with median reaction-energy errors near 0.55–0.60 kcal mol^{−1}. Native GAPW-XC/GTH and GauXC GPW-GTH reach comparable benchmark accuracy on their 65-reaction intersection. Together with the numerical consistency tests, these results validate the native implementation in the molecular regime. The separate 70-reaction comparison of mixed native AE/GTH with GauXC AE/def2-ECP and PySCF spans the UZH and def2 basis/core protocols. Its remaining differences are consistent with the different basis, ECP, and numerical integration settings.

Skala with D3(BJ) achieves excellent agreement with DMC for the three molecular crystals when the AE basis is enlarged uniformly to QZVPP. The CO₂, NH₃, and urea lattice-energy residuals are 2.67, 0.02, and 1.93 kJ mol^{−1}, respectively, giving an MAE of 1.54 kJ mol^{−1} instead of 7.95 kJ mol^{−1} with the matched triple-zeta AE basis. The twelve-phase ice comparison independently reaches MAEs of 1.23 kJ mol^{−1} for absolute lattice energies and 0.77 kJ mol^{−1} for relative energies to Ih. Together, these results identify basis flexibility as a central requirement for assessing the functional’s condensed-phase accuracy. AE and GTH orbital spaces have different core-resolution and optimization requirements, so their nominal zeta levels are not interchangeable accuracy standards. The retained triple-zeta GTH calculations compare competitively with established dispersion-inclusive approximations for ammonia and urea. The ten-solid analysis remains a separate finite-basis structural comparison, with a material-dependent response to basis enlargement in the Si, diamond, and MgO controls. Together with the numerical controls, the molecular-crystal and ice results support both the native implementation and accurate transfer beyond the gas phase, while motivating broader condensed-phase validation with many-body references.

Periodic coupled-cluster and QMC calculations, including FCIQMC where feasible, offer a route to the required many-body reference data. Such reference data would allow the observed transfer limitations to be addressed and tested directly in condensed-phase environments. The native GPW/GAPW implementation makes that development pos-

sible within the electronic-structure framework in which the resulting functional will ultimately be applied.

Conflicts of interest

The authors’ conflict-of-interest declaration is awaiting confirmation.

Data availability

The supporting reaction energies, crystal and molecular energies, equation-of-state data, and numerical controls are included in the ESI. Geometries, executed input files, analysis scripts, and source-version metadata are collected in the native-grid Skala benchmark repository at <https://github.com/DCM-Uni-Paderborn/native-grid-skala-benchmarks>. ESI Sec. S10 identifies the molecular, molecular-crystal, and ten-solid datasets separately and specifies their provenance and numerical limits. A versioned public release of the complete article dataset will accompany submission.

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